

DIGITAL EGYPT PIONEERS INITIATIVE (DEPI)
Data Analysis Diploma



وزارة التخطيط
والتكنولوجيا المعلومات



GRADUATION PROJECT PROPOSAL

Pharmaceutical Market Analysis in Egypt Using IMS Sales Data (2021 – January 2026)

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1. PROJECT TITLE

Pharmaceutical Market Analysis in Egypt Using IMS Sales Data (2021 – January 2026)

Program: Digital Egypt Pioneers Initiative (DEPI) – Data Analysis Diploma

Domain: Pharmaceutical Market Intelligence | Business Intelligence | Predictive Analytics

Data Source: IMS Health / IQVIA Pharmaceutical Sales Data – Egypt (2021–Jan 2026)

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2. EXECUTIVE SUMMARY

The Egyptian pharmaceutical market is one of the most significant and rapidly evolving healthcare markets on the African continent, consistently ranked among the top ten pharmaceutical markets in the Middle East and Africa (MEA) region. With a population exceeding 105 million people, rising chronic disease burden, government-driven healthcare reforms under Egypt Vision 2030, and increasing access to medical insurance, demand for pharmaceuticals — both branded and generic — continues to expand at substantial rates. Despite this growth momentum, the ability of stakeholders to systematically interpret, quantify, and leverage pharmaceutical sales data remains a critical competency gap in the industry.

This graduation project proposes a comprehensive, end-to-end Pharmaceutical Market Analysis of the Egyptian market using five years of IMS (Intercontinental Marketing Services) / IQVIA sales data spanning January 2021 through January 2026. IMS data represents the gold standard for pharmaceutical market intelligence globally, capturing granular transactional information across therapeutic classes, corporations, manufacturers, products, formulations, and market sectors. This dataset — encompassing 22 structured fields including ATC therapeutic classification hierarchies (ATC1–ATC4), NFC formulation codes (NFC1–NFC3), sales values in local currency (LC Value), unit volumes, Moving Annual Totals (MAT), Year-to-Date (YTD) figures, retail pricing, and product launch timelines — provides an exceptional foundation for robust, multi-dimensional analytical work.

The project will deploy a full-spectrum analytical toolkit comprising Microsoft Excel and Power BI for initial exploration and cleaning process, Power BI and Tableau for the development of a professional, interactive Five-pages business intelligence dashboard. The outcome is a rigorous, publication-quality analytical project that mirrors the workflow of a professional pharmaceutical market analyst or business intelligence consultant working within the industry.

The project directly addresses the needs of pharmaceutical companies, sales and marketing directors, business development teams, regulatory affairs professionals, and healthcare policymakers who require evidence-based market intelligence to guide strategic decisions — including portfolio prioritization, market entry planning, competitive response, pricing strategy, and resource allocation. By transforming raw IMS transactional data into actionable intelligence, this project demonstrates the transformative power of data analytics applied to a high-stakes, high-impact real-world domain.

3. PROBLEM STATEMENT

Despite the availability of IMS pharmaceutical sales data, the Egyptian pharmaceutical industry faces a persistent and consequential challenge: the raw data, in its unprocessed form, does not inherently yield actionable intelligence. Companies receive periodic IMS data reports, yet few possess the internal capability to perform the level of granular analytical processing required to translate that data into competitive advantage. This project identifies and addresses the following structural challenges:

3.1 Fragmented Understanding of Market Trends

Pharmaceutical markets are inherently dynamic, influenced by regulatory changes, new product launches, patent expirations, pricing pressures, and seasonal demand fluctuations. Without systematic trend analysis across multiple years and quarters, sales managers and marketing teams cannot reliably distinguish structural market growth from cyclical or temporary performance variations. The absence of longitudinal analysis — comparing 2021 performance through the full 2021–2026 window — leaves decision-makers without the temporal context required for accurate forecasting and strategic planning.

3.2 Inadequate Competitive Intelligence

The Egyptian pharmaceutical market is served by hundreds of corporations and manufacturers, ranging from multinational innovators to domestic generic producers. Determining true market share, identifying share-gainers versus share-losers, and understanding the competitive dynamics within specific therapeutic classes requires rigorous, multi-dimensional comparative analysis across Corporation, Manufacturer, and Product dimensions — simultaneously and over time. Current manual or spreadsheet-based approaches are insufficient for this level of analytical complexity.

3.3 Therapeutic Class Opportunity Blindness

The ATC (Anatomical Therapeutic Chemical) classification system organizes pharmaceuticals into a four-level hierarchy that enables highly structured therapeutic analysis. Without systematic exploitation of ATC1 through ATC4 dimensions, organizations cannot identify which therapeutic areas are growing fastest, which are contracting, and where unmet patient needs represent unaddressed commercial opportunities. This analytical gap leads to misallocation of promotional resources and missed market entry opportunities.

3.4 Pricing Strategy Disconnected from Market Evidence

Retail pricing data within IMS datasets represents a critical but frequently underutilized variable. The relationship between retail price, sales value, and unit volume is complex — particularly in a market where both branded and generic products compete across multiple price tiers. Without price-volume elasticity analysis and price positioning benchmarking, commercial teams cannot make evidence-based pricing decisions or assess the market impact of price changes.

3.5 Deficient Launch Performance Monitoring

Product launches represent significant capital investment and strategic risk. The IMS dataset contains launch date and launches year information for each product, yet systematic performance tracking of new launches — comparing actual sales ramp-up against market benchmarks — is rarely performed with

analytical rigor. This gap prevents organizations from making timely intervention decisions when launches underperform, and from replicating the success drivers of high-performing launches.

3.6 Absence of Data-Driven Forecasting Capability

Sales forecasting in the pharmaceutical industry is often based on qualitative judgement, historical averages, or simple extrapolation. The availability of five years of granular IMS data — including MAT trends, YTD progression, and quarterly patterns — creates a compelling opportunity to build statistically grounded forecasting models. Without such models, organizations remain reactive rather than predictive in their market strategies.

4. PROJECT OBJECTIVES

This project is built around Five core dashboard objectives and two supporting analytical objectives, each aligned with a dedicated Power BI page or analytical capability. Every objective is designed to generate measurable business value for pharmaceutical commercial, marketing, and strategy stakeholders.

Objective 1: Executive Market Overview and Trend Monitoring

Develop the Market Overview Dashboard as the executive entry point for the Egyptian pharmaceutical market intelligence platform. The page displays Total Market Value (LC), Total Unit Volume, and 5-Year CAGR (Units) through KPI cards, supported by MAT trend lines covering 2021–2026. Additional visuals include a Public vs. Private sector donut chart and a Units-by-ATC1 bar chart.

Success Criterion: All universal slicers (Governorate, Launch Year, Corporation, Product, and ATC1 Class) dynamically filter every visual, with KPI updates rendered within 2 seconds.

Objective 2: Geographic Market Demand Analysis

Implement an interactive Azure Map to visualize pharmaceutical demand across Egyptian governorates using unit volume and regional segmentation. A dedicated tooltip page displays market-leading corporation, average price per unit, market share %, total units, and total value for each territory.

Success Criterion: Selecting any governorate filters all dashboard visuals correctly, while tooltip metrics render accurately without errors.

Objective 3: Therapeutic Class Performance Analysis

Build the Therapeutic Dashboard using the ATC hierarchy to evaluate therapeutic market performance. The dashboard includes an ATC2 tree map, dosage form analysis, annual value and unit trends, and a product-level pivot table showing contribution %, growth, and competitive rank.

Success Criterion: The ATC1 slicer dynamically controls all visuals, and tree map tooltips display contextual metrics instantly on hover.

Objective 4: Corporation Competitive Intelligence

Develop the Corporation Performance Dashboard to evaluate pharmaceutical corporations across LC Value and unit volume dimensions. Key visuals include growth contribution waterfalls, corporation growth rankings, market share analysis, and annual trend comparisons. Dynamic KPI cards identify the Market Leader, Fastest Growing Corporation, and Underperforming Corporation.

Success Criterion: Corporation filters update all visuals and KPI calculations dynamically without page refresh.

Objective 5: Product Portfolio Performance and Lifecycle Analysis

Create the Product Performance Dashboard to provide actionable product-level insights for brand and portfolio managers. The dashboard includes Top-7 Product Dominance analysis, Product Lifecycle segmentation, top-performing products by value and units, and a detailed pivot table with pricing, growth, market share, and competitive rank metrics.

Success Criterion: Product rankings remain accurate within the active filter context, and KPI cards update within 3 seconds of slicer interaction.

Objective 6: Manufacturer Performance and Supply Intelligence

Build the Manufacturer Performance Dashboard to analyze manufacturer contribution, growth, and portfolio breadth. Key visuals include manufacturer tree maps, growth contribution waterfalls, Top-10 manufacturers by LC Value, and dosage-form breakdowns. KPI cards identify the Largest Manufacturer and market share contribution.

Success Criterion: Growth contribution analysis correctly distinguishes growth-driving versus growth-diluting manufacturers, with tooltip pages displaying key manufacturer insights accurately.

Objective 7: Cross-Page Filtering and Navigation Architecture

Implement a unified navigation and filtering framework across all dashboard pages. Universal slicers must function consistently across every page, while navigation buttons provide seamless page transitions through a centralized Home page. Custom tooltip pages enhance map, treemap, and product-level interactions.

Success Criterion: Any slicer combination produces a fully synchronized analytical view without rendering issues or cross-filter conflicts.

Objective 8: Advanced DAX Measure Development

Develop a centralized library of advanced DAX measures across dedicated measure tables to support KPIs, rankings, growth analysis, and market share calculations. Measures include Total Value, Total Units, CAGR, Growth %, Market Share %, Product Rank, Corporation Performance Metrics, Manufacturer Contribution, and Average Price Per Unit.

Success Criterion: All measures deliver accurate filter-context-sensitive results with zero circular dependency or blank-output errors.

4A. KEY BUSINESS INSIGHTS

The following executive insights are derived directly from the analytical structure, KPIs, and visual intelligence embedded within the Power BI dashboard. These insights are presentation-ready and designed to support stakeholder briefings, board-level reporting, and strategic planning sessions.

Market Performance and Growth Trajectory

- The Egyptian pharmaceutical market demonstrates sustained multi-year expansion in both LC Value and unit volume as captured by the Market Overview annual trend lines, with MAT-based growth trajectories confirming structural rather than cyclical demand drivers across the 2021–2026 window.
- The 5-Year CAGR (Units) KPI card on the Market Overview page serves as a critical benchmark for distinguishing volume-led growth from value-led growth — a divergence that signals either price inflation outpacing demand, or genuine volumetric market expansion. Strategic planning teams should treat this metric as the primary indicator of market health versus inflationary distortion.
- The Public vs. Private sector donut chart on the Market Overview page reveals the channel dependency profile of the Egyptian pharmaceutical market. A market with elevated Public sector unit share signals significant government procurement concentration, exposing commercial players to tender policy risk — a strategic vulnerability requiring active channel diversification planning.

Competitive Landscape and Corporation Intelligence

- The Corporation Performance Dashboard's Top-5 Market Share donut chart reveals the degree of market concentration in the Egyptian pharmaceutical sector. High concentration among a small cluster of corporations signals barriers to entry and pricing power advantages for incumbents, while a more fragmented distribution indicates active competitive contestability and opportunity for portfolio insurgents.
- The Corporate Growth Contribution waterfall chart distinguishes growth-driving corporations from growth-diluting ones at the market level. Corporations with negative waterfall contributions despite positive overall market growth are losing share — an early warning signal that should trigger competitive response analysis and portfolio reassessment by their commercial leadership teams.
- The Fastest Growing Corporation KPI card identifies the entity capturing disproportionate market share gains relative to the overall market growth rate. This corporation represents the competitive benchmark that all other market participants should measure their performance against — and potentially investigate for portfolio strategy, pricing, or distribution advantages that are driving their outperformance.

Therapeutic Class and Product Portfolio Insights

- The ATC2-level tree map on the Therapeutic Dashboard exposes therapeutic class concentration: segments with disproportionately large tree map tiles relative to their growth rate represent saturated, highly competitive spaces where incremental share gains require significant promotional investment. Conversely, smaller but faster-growing ATC2 segments represent high-return, lower-competition opportunity zones for portfolio expansion and in-licensing consideration.
- The Product Lifecycle donut chart on the Product Performance Dashboard reveals the maturity profile of the active product portfolio. A portfolio skewed toward mature or declining lifecycle stages signals an urgent need for new product launches or in-licensing activity to replenish the growth pipeline. Brand teams should use this visual to quantify the proportion of revenue at risk from portfolio aging within a defined planning horizon.

- The Top-7 Product Dominance funnel chart provides immediate visibility into market share concentration at the product level. In markets where the top 7 products account for a disproportionate share of total units, the remaining portfolio represents a long tail of low-volume assets — many of which may be candidates for rationalization, portfolio pruning, or focused promotional uplift to achieve breakeven contribution thresholds.
- The product pivot table's Product Age (Years) column, combined with Growth % and Market Share %, enables identification of older products that are still growing — a counter-intuitive but commercially significant finding that typically signals strong brand equity, lack of generic competition, or recent label extensions. These products deserve priority investment to sustain their above-lifecycle-average performance.

Geographic and Regional Market Intelligence

- The Azure Map on the Market Overview page enables geographic demand analysis at the governorate level — a capability that is typically absent from standard IMS reporting outputs. Governorates with high unit volumes but low average price per unit (visible via map tooltip) represent high-volume, price-sensitive markets where generic penetration is likely highest and promotional investment ROI may be lower than in premium urban markets.
- The Market Share Units % by Governorate measure, surfaced in the map tooltip, enables field force directors to identify governorates where their selected corporation or product holds above-market-average share versus territories where share is below expectation — directly informing territory investment prioritization and field force deployment decisions at a level of geographic granularity not previously available through standard reporting.

Manufacturer Intelligence and Supply Chain Considerations

- The Top-7 Manufacturer by Product Count tree map reveals portfolio breadth at the manufacturer level. Manufacturers with high product counts but relatively lower total LC Value per product are operating across a broad but shallow portfolio — a potential vulnerability in markets where niche therapeutic expertise and depth, rather than breadth, drives commercial success and premium pricing power.
- The Manufacturer Growth Contribution waterfall chart enables supply chain and business development teams to identify which manufacturers are driving incremental market value growth versus which are experiencing revenue contraction. Manufacturers with negative growth contribution despite positive market growth are losing competitive relevance — potentially signaling product quality issues, pricing disadvantages, supply constraints, or failure to launch new products at the pace required to maintain market position.
- The Largest Manufacturer Share % KPI card, combined with the Top-10 Manufacturer by Value bar chart, enables assessment of supply chain concentration risk. Markets where one or two manufacturers account for a dominant share of total value indicate systemic supply dependency — a strategic risk that procurement teams, regulatory affairs functions, and market access directors must incorporate into business continuity planning and sourcing diversification strategies.

Dosage Form, Pricing, and Route-of-Administration Dynamics

- The Route-of-Administration bar charts available on both the Market Overview page (Dosage Forms by Units) and the Therapeutic and Manufacturer Dashboard pages enable formulation-level demand analysis. Dominant oral dosage form volumes confirm the accessibility-driven nature of the Egyptian retail pharmacy market, while elevated injectable or speciality formulation volumes within specific ATC classes signal therapeutic complexity and potentially higher price-per-unit opportunities for manufacturers with advanced formulation capabilities.
- The Average Price Per Unit measure, surfaced across multiple tooltip pages and the Product pivot table's Retail Price column, enables price positioning analysis at the product, therapeutic class, and geographic level. Markets or therapeutic segments where Average Price Per Unit

significantly exceeds the portfolio average represent premium positioning opportunities — provided that volume levels are sufficient to justify the commercial infrastructure required to maintain that positioning against generic competition.

5. SCOPE OF WORK

5.1 Temporal Scope

The analytical scope spans the five-year period from January 2021 through January 2026, encompassing 20+ calendar quarters of granular pharmaceutical sales data. This temporal range is sufficient to capture full business cycles, identify multi-year trends, observe the impact of macro-economic events (including post-COVID market recovery dynamics in Egypt), and build statistically meaningful forecasting models.

5.2 Market Scope

The analysis covers the total Egyptian pharmaceutical market as captured by IMS / IQVIA data, segmented by:

- Market Sector: Public (government/tender) and Private (retail pharmacy) channels
- Therapeutic Classification: All ATC1 anatomical groups through to ATC4 chemical substance level
- Formulation Categories: All NFC1 through NFC3 dosage form classifications
- Corporation: All pharmaceutical corporations active in the Egyptian market
- Manufacturer: All manufacturers (innovator and generic) supplying the Egyptian market
- Product: Individual branded and generic pharmaceutical products with unique Pack and Strength identifiers

5.3 Analytical Scope

The analytical work encompasses the following domains:

- Descriptive Analytics: Historical performance characterization and trend identification
- Comparative Analytics: Multi-dimensional benchmarking across corporations, products, and therapeutic classes
- Prescriptive Insights: Evidence-based recommendations for market strategy and portfolio decisions

5.4 Out of Scope

The following elements are explicitly excluded from this project's scope: patient-level clinical data, primary market research or surveys, regulatory submissions or approval processes, financial modelling beyond sales performance metrics, and markets outside Egypt.

6. KEY KPIS AND METRICS

The following table presents the complete KPI framework for this project, including definitions, data sources, and business relevance. All KPIs are derived from the 22 available dataset columns and are designed to support both strategic and operational pharmaceutical market intelligence needs.

KPI / Metric	Definition	Data Column(s) Used	Business Value
Total Sales Value (LC)	Aggregate local-currency revenue across all products and periods	LC Value, Period	Measures overall market size and revenue trajectory
Total Units Sold	Sum of pack units sold across all SKUs and time periods	Units, Period, Calendar Year	Reflects volume-based market penetration and demand
Market Share %	Corporation or product revenue as % of total market LC Value	LC Value, Corporation, Product	Benchmarks competitive positioning at product/company level
Growth Rate % (Value & Units)	YoY or QoQ percentage change in sales value or units	LC Value, Units, Calendar Year, Calendar Quarter	Identifies high-growth and declining market segments
Moving Annual Total (MAT)	Rolling 12-month cumulative sales; smooths seasonality	Moving Annual Total	Provides trend direction independent of seasonal variation
Year-to-Date (YTD) Performance	Cumulative sales from fiscal year start to current period	Year to Date, Calendar Year	Enables mid-year tracking vs. annual targets
Top Products Ranking	Products ranked by LC Value and Units within defined periods	Product, LC Value, Units	Identifies blockbuster and declining brands
Top Corporations Ranking	Corporations ranked by total market LC Value share	Corporation, LC Value	Reveals competitive landscape among pharma companies
Top Manufacturers Ranking	Manufacturer share of units and value	Manufacturer, Units, LC Value	Distinguishes originator vs. generic manufacturer dynamics
Price vs. Sales Analysis	Correlation between Retail Price and sales volume/value	Retail Price, Units, LC Value, Product	Quantifies price elasticity and premium brand behaviour
New Launch Performance	Sales ramp-up of products post-launch date	Launch Date, Launch Year, LC Value, Units	Evaluates launch success and market acceptance speed
ATC Segment Analysis	Sales breakdown by therapeutic class hierarchy	ATC1–ATC4, LC Value, Units	Reveals therapeutic area opportunities and growth pockets
NFC Formulation Analysis	Sales split by dosage form and formulation category	NFC1–NFC3, Units, LC Value	Informs product mix and formulation preference trends
Sector Performance	Public vs. private sector market split	Sector, LC Value, Units	Guides market access strategy and tender business focus

7. DATA CLEANING AND PREPARATION PROCESS

A rigorous, documented data preparation pipeline is foundational to the analytical integrity of this project. IMS pharmaceutical sales data, while structured and professionally curated, requires systematic validation and transformation before it is suitable for advanced analytics. The following multi-stage process will be applied:

Stage 1: Data Ingestion and Initial Assessment

Raw IMS data will be ingested from its source format (Excel or CSV) into a power query environment. An initial structural assessment will catalogue the number of records, unique values per column, data types, memory footprint, and file integrity. A data profiling report will be generated documenting the baseline state of all 22 columns prior to any transformation.

Stage 2: Missing Value Analysis and Treatment

Each column will be assessed for null, blank, and placeholder values. Missing values in categorical fields (e.g., ATC3, ATC4, NFC3, Manufacturer) will be investigated to determine whether they represent genuine data absences or encoding conventions in IMS reporting. Where appropriate, hierarchical imputation will be applied — for example, deriving ATC3 from ATC2 context. Numerical fields (LC Value, Units, Retail Price) with missing values will be treated through median imputation or forward-fill based on temporal context, with all imputations documented.

Stage 3: Duplicate Detection and Removal

Duplicate records will be identified at multiple levels of granularity: full-row duplicates, duplicate product-period combinations, and logical duplicates arising from minor spelling variations in product or corporation names. Deduplication logic will be validated manually on a random sample of flagged records prior to automated removal.

Stage 4: Data Standardization and Normalization

Standardization will be applied to all categorical text fields to resolve case inconsistencies, leading/trailing whitespace, abbreviation variations, and encoding differences. Corporation and Manufacturer names will be reconciled against a master reference list. ATC and NFC codes will be validated against the WHO ATC classification and NFC standards. Date fields (Launch Date, Calendar Quarter, Period) will be normalized to consistent datetime formats enabling time-series operations.

Stage 5: Data Type Validation and Schema Enforcement

All columns will be enforced to their correct data types: LC Value and Units as numeric float and integer respectively; Launch Year and Calendar Year as integer; Retail Price as float; Launch Date as datetime; all categorical fields as string/category.

Stage 6: Derived Variable Creation

A set of analytically critical derived variables will be engineered from the base fields:

- Market Share %: LC Value of product/corporation as a proportion of total market LC Value per period
- YoY Growth Rate: Percentage change in LC Value and Units between equivalent periods across years

- Launch Age (Months): Calculated from Launch Date to each reporting Period for launch performance curves
- Price Band Category: Retail Price grouped into Low / Mid / Premium tiers using quantile-based segmentation
- MAT Growth: Year-over-year change in Moving Annual Total value and units

Stage 7: Data Quality Documentation

Upon completion of the cleaning pipeline, a comprehensive Data Quality Report will be produced documenting: record counts before and after cleaning, columns affected by each cleaning operation, imputation decisions and their rationale, validation results, and a final data dictionary describing all 22 original columns plus all derived variables with definitions, data types, and value ranges.

8. TOOLS AND TECHNOLOGIES

This project adopts an industry-standard, multi-tool analytical stack that mirrors the technology environment used by professional pharmaceutical market analysts and business intelligence consultants. Each tool is selected for its specific capability contribution and industry relevance:

Tool / Technology	Role in Project	Category
Microsoft Excel	Data cleaning, pivot tables, exploratory analysis, quick charting	Data Preparation
Power BI (Desktop & Service)	Interactive dashboard development, DAX measures, drill-through reports	Business Intelligence
Tableau	Supplementary advanced visualizations, geographic pharma market mapping	Data Visualization
Microsoft PowerPoint	Executive presentation design and stakeholder reporting	Reporting

The integration of these tools across the project workflow creates a seamless pipeline from raw data ingestion to executive-ready reporting — demonstrating the full-stack data analytics competency expected of graduates of the DEPI Data Analysis Diploma program.

9. POWER BI & TABLEAU DASHBOARD DESIGN PLAN

The centerpiece deliverable of this project is a professional, interactive Power BI dashboard comprising Five purpose-built pages, each targeting a distinct analytical audience and decision-making need. The dashboard will be designed to executive presentation standards, with consistent branding, intuitive navigation, dynamic slicers, and drill-through capabilities.

#	Dashboard Page	Key Visuals & Metrics
1	Executive Market Overview	High-level KPI scorecards: total market value, YTD growth, MAT trend, top 5 corporations, market CAGR since 2021
2	Therapeutic Dashboard	Tree map ATC2 + Pivot Table
3	Corporation Performance	Waterfall + Bar (Growth %) + Donut (Top 5 Market Share) + KPI (Market Leader, Fastest Growing, Underperforming)
4	Product Performance	Funnel (Top 7) + Product Lifecycle Donut + Top 10 + Pivot Table
5	Manufacturer Performance	Tree map + Waterfall + Bar Chart + KPI Cards

9.1 Dashboard Design Principles

- Corporate Color Scheme: Consistent professional color palette aligned with pharmaceutical industry standards
- Navigation: Bookmarked page navigation panel visible on all dashboard pages
- Dynamic Filtering: Universal slicers for Year, Quarter, ATC1 Class, Corporation, Sector, and Period
- KPI Cards: Prominent scorecards displaying current period performance vs. prior period and target
- Drill-Through: Right-click drill-through from market overview to product and corporation detail pages
- Mobile Optimization: Mobile-responsive layout for key KPI views
- Tooltips: Custom tooltip pages providing contextual data on all visual elements
- DAX Measures: Custom DAX calculations for MAT, YTD, Market Share %, Growth Rates, and Rankings

10. EXPECTED BUSINESS IMPACT

This project delivers substantial and measurable business value across multiple stakeholder categories within the Egyptian pharmaceutical industry. The analytical outputs produced directly support commercial decision-making at the strategic, tactical, and operational levels:

10.1 For Pharmaceutical Companies and Commercial Directors

The market growth analysis and competitive benchmarking outputs provide pharmaceutical companies with a clear, evidence-based picture of where the Egyptian market is growing, where competitive intensity is highest, and where strategic investment should be prioritized. The ability to quantify market share evolution over five years, identify share-gaining corporations, and detect early signals of market disruption

transforms what would otherwise be qualitative commercial judgement into data-driven strategic decision-making. Companies can use the MAT trend analysis and forecasting dashboard to set scientifically grounded annual sales targets and resource allocation plans.

10.2 For Sales Management and Regional Teams

The product and corporation performance dashboards provide sales managers with granular, real-time-comparable performance data at the product and therapeutic class level. The ability to rapidly identify which products are overperforming or underperforming relative to their market benchmarks — and to decompose performance by sector, territory proxy, and period — enables more agile and informed sales force deployment decisions. YTD tracking against MAT benchmarks supports monthly performance reviews and early-warning systems for sales shortfalls.

10.3 For Marketing and Brand Teams

The ATC therapeutic class analysis and new launch performance tracking outputs deliver actionable intelligence for brand strategy development. Marketing teams can identify which therapeutic segments are experiencing the strongest patient demand growth, benchmark their brand's growth against the competitive set within the same ATC class, and evaluate the effectiveness of product launch strategies through objective performance curves. The price-volume analysis supports evidence-based pricing decisions and premium brand positioning.

10.4 For Business Development and Portfolio Strategy

The market gap identification and growth opportunity matrix outputs support business development teams in identifying acquisition targets, licensing opportunities, and in-licensing priorities. By mapping therapeutic class growth rates against current corporate presence, the analysis reveals where portfolio gaps exist and where the Egyptian market offers the highest risk-adjusted growth potential for new asset development or in-licensing.

10.5 For Healthcare Policymakers and Regulatory Affairs

The sector-level analysis (public vs. private) and therapeutic class penetration data provide healthcare policymakers with evidence-based insights into access gaps, market concentration risks, and the effectiveness of national medicine pricing and reimbursement policies. The longitudinal analysis of new product launches by therapeutic class illuminates the pace of market access for innovative medicines in Egypt.

10.6 Contribution to DEPI Program Objectives

Beyond its pharmaceutical industry applications, this project serves as a demonstration of the transformative potential of data analytics education under the Digital Egypt Pioneers Initiative. It showcases how DEPI graduates can apply advanced analytical skills to solve complex, high-value problems in Egypt's priority economic sectors — directly contributing to the programmer's goal of building a world-class digital workforce capable of driving Egypt's knowledge economy.

11. PROJECT TIMELINE

The project is structured across eight sequential phases spanning sixteen weeks. This timeline is designed to ensure rigorous analytical quality while maintaining realistic delivery milestones aligned with DEPI program requirements. Each phase produces a documented output that serves as input for the subsequent phase.

Phase	Activity	Timeline	Key Deliverable
Phase 1	Project Initiation & Data Acquisition	Week 1	Project scope document, raw dataset validated and catalogued
Phase 2	Data Cleaning & Preparation	Week 2	Cleaned, standardized dataset; data quality report; SQL schema
Phase 3	Power BI & Tableau Dashboard Development	Week 3	Six-page interactive Power BI dashboard (PBIX file)
Phase 4	Reporting & Documentation	Week 4	Full analytical report (Word), executive presentation (PowerPoint)
Phase 5	Review, QA & Final Submission	Week 5	Final deliverables package, project demo, submission-ready proposal

Total Project Duration: 5 Weeks | Milestones reviewed at the end of each phase by the project supervisor. Phase deliverables are submitted for progressive assessment, enabling early identification of scope adjustments or analytical challenges.

12. EXPECTED DELIVERABLES

Upon successful completion of this project, the following eight formal deliverables will be submitted to the DEPI program supervisory committee. Each deliverable represents a professional-grade output aligned with industry standards for pharmaceutical market intelligence work:

ID	Deliverable	Format	Description
D1	Cleaned & Validated IMS Dataset	Excel / CSV	Fully processed dataset with documented cleaning log and data dictionary
D2	Power BI Interactive Dashboard	PBIX File	Six-page interactive dashboard with slicers, drill-throughs, and bookmarks
D3	Tableau Interactive Dashboard	twb File	Six-page interactive dashboard with slicers, drill-throughs, and bookmarks
D4	Executive Presentation Deck	MS PowerPoint (.pptx)	Professional 20–25 slide deck for stakeholder presentation and project defence
D5	Project Proposal Document	MS Word (.docx)	This document – structured academic proposal approved by supervisory committee
D6	Final Project Documentation	PDF Package	Complete project portfolio including all deliverables, methodology, and references

All deliverables will be version-controlled, thoroughly documented with methodology notes, and submitted as a consolidated final project package. The Power BI dashboard will be published to the Power BI Service for live demonstration during the project defense session.

13. CONCLUSION

This project proposal presents a rigorous, analytically ambitious, and commercially grounded graduation project that fully leverages the breadth and depth of the Digital Egypt Pioneers Initiative Data Analysis Diploma curriculum. By applying the full spectrum of data analysis tools and techniques — from foundational data cleaning and SQL engineering through advanced Python statistical modelling and professional Power BI dashboard development — to one of the most commercially significant datasets available in the Egyptian business landscape, this project delivers both exceptional academic merit and genuine real-world applicability.

The Egyptian pharmaceutical market represents a dynamic, high-stakes analytical environment where data-driven decision-making has the capacity to redirect hundreds of millions of Egyptian pounds in commercial investment, optimize patient access to essential medicines, and inform national healthcare policy. By transforming five years of granular IMS sales data into a comprehensive, interactive, and actionable intelligence platform, this project demonstrates precisely the caliber of analytical capability that the DEPI program is designed to cultivate and certify.

The eight analytical objectives, fourteen KPI dimensions, six-dashboard Power BI deliverable, and integrated forecasting capability collectively position this project as one of the most comprehensive and impactful graduation projects within the pharmaceutical and healthcare analytics domain. The project is feasible within the proposed sixteen-week timeline, grounded in real data with genuine commercial provenance, and structured to produce deliverables of direct value to industry practitioners.

It is respectfully submitted for the consideration and approval of the DEPI program supervisory committee, with full confidence that it meets and exceeds the standard expected for graduation project approval.